ITA Classroom Activity 2

Date: 18.07.2025 (Friday)

Hours: 1 & 2 Hours

1. Context:

Mawsynram and Cherrapunji, both located in Meghalaya, India, are known for receiving extremely high rainfall. Suppose a weather monitoring system is trying to **predict the daily rainfall pattern** in both places during the monsoon season.

You are given the following:

Rainfall Level	Probability in Mawsynram	Probability in Cherrapunji
No Rain	0.05	0.10
Light Rain	0.10	0.20
Moderate Rain	0.35	0.40
Heavy Rain	0.50	0.30

(a) Compute the **information content (in bits)** of each rainfall category for both Mawsynram and Cherrapunji using Hartley's formula:

$$\mathbf{I(x) = -\log_2(p(x))}$$

(b) Compute the **Shannon entropy H(X)** for both places:

$$\mathbf{H(X) = -\Sigma\, p(x) \log_2(p(x))}$$

(c) Interpret the results:
Which place has higher entropy?
What does that mean about the **uncertainty** or **predictability** of rainfall in each region?

(d) If the system needs to transmit the **daily rainfall pattern** of both places over a satellite link, **how many bits per day** and **how many bits per week** would it approximately require on average?

(a) Information Content (in bits) for Each Rainfall Category
We use Hartley's formula:

$$I(x) = -\log_2(p(x))$$

Mawsynram:

- No Rain $p = 0.05$: $I(x) = -\log_2(0.05) = 4.32$ bits
- Light Rain $p = 0.10$: $I(x) = -\log_2(0.10) = 3.32$ bits
- Moderate Rain $p = 0.35$: $I(x) = -\log_2(0.35) = 1.51$ bits
- Heavy Rain $p = 0.50$: $I(x) = -\log_2(0.50) = 1.00$ bit

Cherrapunji:

- No Rain $p = 0.10$: $I(x) = -\log_2(0.10) = 3.32$ bits
- Light Rain $p = 0.20$: $I(x) = -\log_2(0.20) = 2.32$ bits
- Moderate Rain $p = 0.40$: $I(x) = -\log_2(0.40) = 1.32$ bits
- Heavy Rain $p = 0.30$: $I(x) = -\log_2(0.30) = 1.74$ bits

(b) Shannon Entropy $H(X)$

$$H(X) = -\sum p(x) \log_2(p(x))$$

Mawsynram:

$$\begin{aligned} H(X) &= [0.05 \times 4.32] + [0.10 \times 3.32] + [0.35 \times 1.51] + [0.50 \times 1.00] \\ &= (0.216) + (0.332) + (0.529) + (0.500) \\ &= 1.577 \text{ bits} \end{aligned}$$

Cherrapunji:

$$\begin{aligned} H(X) &= [0.10 \times 3.32] + [0.20 \times 2.32] + [0.40 \times 1.32] + [0.30 \times 1.74] \\ &= (0.332) + (0.464) + (0.528) + (0.522) \\ &= 1.846 \text{ bits} \end{aligned}$$

(c) Interpretation

- **Cherrapunji has higher entropy (1.85 bits) vs Mawsynram (1.58 bits).**
- **Higher entropy means higher uncertainty**: The rainfall in Cherrapunji is *less predictable* (more random) than in Mawsynram.
- **Lower entropy (Mawsynram)** implies more predictability—the rainfall pattern is more consistent.

(d) Bits Required to Transmit Daily Pattern
For both places (per day):

(d) Bits Required to Transmit Daily Pattern
For both places (per day):
Mawsynram: 1.58 bits/day
Cherrapunji: 1.85 bits/day
Total per day:
1.58
+

ITA Classroom Activity 3

2. Simulate a dataset of toss results for three recent ODI captains from **India**, **Australia**, and **England**, assuming each has participated in **12 tosses**. Calculate entropy and analyze the uncertainty in outcomes:

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Answers Summarized

(a) Information content (bits) for each rainfall category:

- Mawsynram: 4.32, 3.32, 1.51, 1.00 bits (No, Light, Moderate, Heavy)
- Cherrapunji: 3.32, 2.32, 1.32, 1.74 bits (No, Light, Moderate, Heavy)

(b) Shannon Entropy:

- Mawsynram: **1.58 bits**
- Cherrapunji: **1.85 bits**

(c) Which place has higher entropy?

- **Cherrapunji** (thus, rainfall is less predictable there)

(d) Bits required to transmit the daily rainfall pattern:

- **Per day: 3.43 bits**
- **Per week: 24 bits (rounded to nearest integer)**